CLASS 6 — MATHS · FORMULA SHEET

Chapter 6: **Perimeter and Area** | Quick revision, one page

Perimeter

- Perimeter = total length of the boundary. Units: cm, m, km.
- Rectangle: $P = 2(l + b)$. Square: $P = 4a$.
- Triangle: $P = a + b + c$ (sum of three sides). Any polygon: sum of all sides.

Area

- Area = the surface a shape covers. Units: cm^2, m^2, km^2.
- Rectangle: $A = l \times b$. Square: $A = a \times a = a^2$.
- Triangle: $A = \frac{1}{2} \times \text{base} \times \text{height}$ (half of the rectangle on the same base and height).
- Finding a side: $b = \frac{A}{l}$ for a rectangle; $a = \frac{P}{4}$ for a square.

Units

- $1\ \text{m} = 100\ \text{cm}$, so $1\ \text{m}^2 = 100 \times 100 = 10000\ \text{cm}^2$.
- Cost of fencing = perimeter × rate per metre. Cost of paving/tiling = area × rate per m^2.
- Number of tiles = area of floor ÷ area of one tile.

Composite shapes

- Split an L-shaped or odd shape into rectangles, find each area, and add; or subtract the cut-out part from the big rectangle.
- Same perimeter does not mean same area.

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